# Binomial Theorem

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# Binomial Theorem

## INTRODUCTION

Binomial is made of "Bi" which means "two" and "nomial" which means an expression with numbers or variables.

### Binomial Expression

eg : $x + y$, $yz + \frac{1}{x}$

**Note :**

$x + (y + z)$ is also a binomial expression where 'x' is first term and '$(y + z)$' is second term. Similarly, $(x + y) + (z + w)$ is a binomial expression where $(y + z)$ is first term and $(z + w)$ is second term.

**Definition**

An algebraic expression which contains two dissimilar terms is called **binomial expression**.

### Binomial Theorem

$(x + y)^2 = x^2 + y^2 + 2xy$

$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$

$(x + y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$

When index is very high then use binomial theorem as below

$$(x + y)^n = {}^nC_0x^ny^0 + {}^nC_1x^{n-1}y^1 + {}^nC_2x^{n-2}y^2 + \ldots + {}^nC_rx^{n-r}y^r + \ldots + {}^nC_nx^{n-n}y^n$$

Where $n \in N$, x and y $\in$ set of complex numbers and ${}^nC_0$, ${}^nC_1$, ${}^nC_2$... are called binomial coefficients normally written as $C_0$, $C_1$, $C_2$... (n hidden)

**Definition**

The formula by which any positive integral power of a binomial expression can be expanded in the form of a series is known as **Binomial Theorem**.

### Proof of Binomial Expansion

$(x + y)^n = (x + y)(x + y)\ldots(x + y)$ {n brackets}

Each term in the expansion is formed by taking one letter from each bracket and multiplying them together. For example, choose x from all the bracket and multiplying them, $x^n$ is obtained. This can be done in 1 (= ${}^nC_0$) way. Now, choose x from $(n - 1)$ brackets and y from remaining one bracket and multiplying them, $x^{n-1}y^1$ is obtained. But number of ways of choosing $(n - 1)$ brackets out of 'n' is ${}^nC_{n-1} = {}^nC_1$ ways.

**Points to Remember!!!**

**(i)** The number of terms in the expansion is $(n + 1)$
i.e. one more than the index (n)

**(ii)** $T_{r+1} = {}^nC_rx^{n-r}y^r$ is called the general term of the binomial expansion.

**(iii)** Sum of powers of each term in $(x + y)^n$ is n.

Hence ${}^nC_1 \times {}^1C_1 = {}^nC_1 = {}^nC_1$ terms of the form $x^{n-1}y^1$ are generated. Continuing in this way, ${}^nC_r$ terms of the form $x^{n-r}y^r$ are generated & the process is done till all possible combinations of powers of 'x' and 'y' are obtained. Adding all $(x + y)^n = x^n + {}^nC_1x^{n-1}y^1 + ... + {}^nC_rx^{n-r}y^r + ... + y^n$

## Highlights

$$(x+y)^n = \sum_{r=0}^{n} {}^nC_r x^{n-r}y^r = \sum_{r=0}^{n} T_{r+1}$$

Where $T_{r+1}$ is general term

Also, $(x+y)^n = (y+x)^n = \sum_{r=0}^{n} {}^nC_r y^{n-r}x^r$

## $p^{th}$ term from the end

- $p^{th}$ term from the end in the expansion of $(x + y)^n$ is same as $(n - p + 2)^{th}$ term from the beginning of the expansion
- $r^{th}$ term binomial coefficient from the end is ${}^nC_{n-(r-1)} = {}^nC_{n-r+1}$
- $(x + y)^n = {}^nC_0x^ny^0 + {}^nC_1x^{n-1}y^1 + {}^nC_2x^{n-2}y^2 + ... + {}^nC_{n-2}x^2y^{n-2} + {}^nC_{n-1}x^1y^{n-1} + {}^nC_nx^0y^n$
  Binomial coefficients of terms equidistance from beginning and end are equal i.e.

$${}^nC_0 = {}^nC_n$$

$${}^nC_1 = {}^nC_{n-1}$$

$${}^nC_r = {}^nC_{n-r}$$

## Frequently Used Binomial Expansion

**(a)** $(x + y)^n = {}^nC_0x^ny^0 + {}^nC_1x^{n-1}y^1 + ... + {}^nC_nx^0y^n$

$$= \sum_{r=0}^{n} {}^nC_r x^{n-r}y^r$$

**(b)** $(x - y)^n = {}^nC_0x^ny^0 - {}^nC_1x^{n-1}y^1 + {}^nC_2x^{n-2}y^2 - {}^nC_3x^{n-3}y^3 + ... + (-1)^n {}^nC_nx^0y^n$

$$= \sum_{r=0}^{n} (-1)^r {}^nC_r x^{n-r}y^r$$

**(c)** $(1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + ... + {}^nC_rx^r + ... + {}^nC_nx^n$

$$= \sum_{r=0}^{n} {}^nC_r x^r$$

**(d)** $(1 - x)^n = {}^nC_0 - {}^nC_1x + {}^nC_2x^2 - {}^nC_3x^3 + \ldots + (-1)^r {}^nC_rx^r + \ldots + (-1)^n {}^nC_nx^n$

$$= \sum_{r=0}^{n} (-1)^r {}^nC_r x^r$$

**Points to Remember!!!**

**(1)** Coefficient of $x^r$ in expansion of $(1 + x)^n = {}^nC_r$

**(2)** Coefficient of $x^r$ in expansion of $(1 - x)^n = (-1)^r {}^nC_r$

**Q.** **If the fourth term in the binomial expansion of $\left(\sqrt{x^{\left(\frac{1}{1+\log_{10}x}\right)}} + x^{\frac{1}{12}}\right)^6$ is equal to 200, and $x > 1$, then the value of x is**

**(A)** $10^3$ **(B)** 100 **(C)** $10^4$ **(D)** 10

**Sol.** **(D)**

$${}^6C_3\left[\left(x^{\left(\frac{1}{1+\log_{10}x}\right)}\right)^{\frac{1}{2}}\right]^3\left(x^{\frac{1}{12}}\right)^3 = 200$$

$$x^{\frac{3}{2}\left(\frac{1}{1+\log_{10}x}\right)} \cdot x^{\frac{1}{4}} = 10$$

$$\frac{3}{2}\left(\frac{1}{1+\log_{10}x}\right)\log_{10}x + \frac{1}{4}\log_{10}x = 1 \qquad \text{(Taking log both side)}$$

$$\frac{3}{2}\frac{1}{(1+\alpha)}\alpha + \frac{\alpha}{4} = 1 \Rightarrow \qquad \alpha^2 + 3\alpha - 4 = 0$$

$\alpha = 1, -4$

$\therefore \quad \log_{10}x = \alpha = 1, -4$

$x = 10, 10^{-4}$

**Q.** **Find the coefficient of $x^{15}$ in the expansion of $(x - x^2)^{10}$.**

**Sol.** $[x(1 - x)]^{10} = x^{10}(1 - x)^{10}$

$\therefore$ coeff. of $x^{15}$ in $x^{10}(1 - x)^{10}$ =

coeff. of $x^5$ in $(1 - x)^{10} = (-1)^5 {}^{10}C_5$

$= -252$

**Q.** **Find the fourth term in the expansion of** $\left(2x - \frac{y}{2}\right)^7$

**Sol.** $T_4 = T_{3+1} = {}^7C_3(2x)^{7-3}\left(-\frac{y}{2}\right)^3$

$= \frac{7 \times 6 \times 5}{6} \times 16x^4(-1)\frac{y^3}{8}$

$= -70x^4y^3$

**Q.** **Find term involving $x^3$ in** $\left(2x^2 - \frac{1}{3x}\right)^6$

**Sol.** $T_{r+1} = {}^6C_r\left(2x^2\right)^{6-r}\left(-\frac{1}{3x}\right)^r$

$= {}^6C_r\left(2^{6-r}\right)\frac{x^{12-2r}}{x^r}\left(-\frac{1}{3}\right)^r$

Equating power of x to 3

$12 - 3r = 3 \Rightarrow r = 3$

$\therefore$ Term $= {}^6C_3\left(2x^2\right)^3\left(-\frac{1}{3x}\right)^3 = \frac{-160}{27}x^3$

**Q.** **Coefficient of $x^{50}$ in $(1 + x)^{101}(1 - x + x^2)^{100}$ is-**

**Sol.** $(1 + x)(1 + x)^{100}(1 - x + x^2)^{100} = (1 + x)[1 + x^3]^{100}$
$=$ coeff. of $x^{50}$ in $1.(1 + x^3)^{100} +$ coeff. of $x^{50}$ in $x(1 + x^3)^{100}$
$= 0 + 0 = 0$

**Q.** **Find coefficient of $x^6$ in $(1 + 3x + 3x^2 + x^3)^{15}$**

**Sol.** Given expansion is $[(1 + x)^3]^{15} = (1 + x)^{45}$
$\therefore$ coefficient of $x^6$ in $(1 + x)^{45}$ is ${}^{45}C_6$

**Q.** **Show that coefficient of $x^5$ in the expansion of $(1 + x^2)^5 . (1 + x)^4$ is 60.**

**Sol.** $({}^5C_0 + {}^5C_1x^2 + {}^5C_2x^4 + {}^5C_3x^6 + {}^5C_4x^8 + {}^5C_5x^{10}) \times ({}^4C_0 + {}^4C_1x + {}^4C_2x^2 + {}^4C_3x^3 + {}^4C_4x^4)$

Required coefficient is

${}^5C_1\ {}^4C_3 + {}^5C_2\ {}^4C_1 = 20 + 40 = 60.$

**Q.** **Let $\left(1+x^2\right)^2 .(1+x)^n = \sum_{k=0}^{n+4} a_k x^k$. If $a_1$, $a_2$ and $a_3$ are in A.P. find n.**

**Sol.** $(1 + 2x^2 + x^4)\ (1 + {}^nC_1x + {}^nC_2x^2 + \ldots) =$

$a_0x^0 + a_1x^1 + a_2x^2 + \ldots$

comparing coefficients of like powers of x both side

$a_1 = {}^nC_1,\ a_2 = 2 + {}^nC_2,\ a_3 = {}^nC_3 + 2({}^nC_1)$

$2a_2 = a_1 + a_3$

$$2\left[2+\frac{n(n-1)}{2}\right] = {}^nC_1 + \frac{n(n-1)(n-2)}{6} + 2n$$

$6[4 + n(n - 1)] = 18n + n^3 - 3n^2 + 2n$

$24 + 6n^2 - 6n = 18n + n^3 - 3n^2 + 2n$

$n^3 - 9n^2 + 26n - 24 = 0$

$(n - 2)\ (n - 3)\ (n - 4) = 0 \Rightarrow n = 2, 3, 4$

**Q.** **The coefficient of $(3r)^{th}$ term and coefficient of $(r + 2)^{th}$ term in the expansion of $(1 + x)^{2n}$ are equal ($r > 1$, $n > 2$ and positive integer) then**

**(A)** $r = \frac{n}{2}$ **(B)** $r = \frac{n}{3}$ **(C)** $r = \frac{n+1}{2}$ **(D)** $r = \frac{n-1}{2}$

**Sol.** **(A)**

${}^{2n}C_{3r-1} = {}^{2n}C_{r+1}$

| | | | |
|---|---|---|---|
| $3r - 1 = r + 1$ | or | $2n = 3r - 1 + r + 1$ | $\left({}^nC_x = {}^nC_y \Rightarrow x + y = n \text{ or } x = y\right)$ |
| $2r = 2$ | or | $2n = 4r$ | |
| $r = 1$ | or | $n = 2r$ | |

**Q.** **If in the expansion of $(1 + y)^n$, then coefficient of $5^{th}$, $6^{th}$ and $7^{th}$ terms are in A.P., then 'n' is equal to**

**(A) 7, 11** **(B) 7, 14** **(C) 8, 16** **(D) None**

**Sol.** **(B)**

$2\,{}^{n}C_5 = {}^{n}C_4 + {}^{n}C_6$

$$\frac{2(n!)}{5!(n-5)!} = \frac{n!}{4!(n-4)!} + \frac{n!}{6!(n-6)!}$$

$12(n-4) = 30 + (n-5)(n-4)$

$n^2 - 21n + 98 = 0 \Rightarrow n = 7, 14$

**Q.** **If $\sum_{r=0}^{2n} a_r (x-2)^r = \sum_{r=0}^{2n} b_r (x-3)^r$ and $a_k = 1\ \forall\ k \geq n$, then show that $b_n = {}^{2n+1}C_{n+1}$**

**Sol.** $a_0(x-2)^0 + a_1(x-2)^1 + a_2(x-2)^2 + \ldots + a_{n-1}(x-2)^{n-1} + a_n(x-2)^n + a_{n+1}(x-2)^{n+1} + \ldots + a_{2n}(x-2)^{2n}$

$= b_0(x-3)^0 + b_1(x-3)^1 + b_2(x-3)^2 + \ldots + b_{2n}(x-3)^{2n}$

$a_0 + a_1(x-2)^1 + a_2(x-2)^2 + \ldots + a_{n-1}(x-2)^{n-1} + a_n(x-2)^n + a_{n+1}(x-2)^{n+1} + \ldots + a_{2n}(x-2)^{2n}$

$= b_0 + b_1(x-3)^1 + b_2(x-3)^2 + \ldots + b_n(x-3)^n + b_{n+1}(x-3)^{n+1} + \ldots + b_{2n}(x-3)^{2n}$

Put $x - 3 = t$

$a_0 + a_1(t-1)^1 + a_2(t-1)^2 + \ldots + a_{n-1}(t-1)^{n-1} + a_n(t-1)^n + a_{n+1}(t-1)^{n+1} + \ldots + a_{2n}(t-1)^{2n}$

$= b_0 + b_1 t + b_2 t^2 + \ldots + b_n t^n + b_{n+1} t^{n+1} + \ldots + b_{2n} t^{2n}$

**(M 1)**

Comparing coefficient of $t^n$ from both sides-

${}^{n}C_0 + {}^{n+1}C_1 + {}^{n+2}C_2 + \ldots + {}^{2n}C_n = b_n$

$b_n = ({}^{n+1}C_{n+1} + {}^{n+1}C_n) + {}^{n+2}C_n + \ldots + {}^{2n}C_n$

$= ({}^{n+2}C_{n+1} + {}^{n+2}C_n) + \ldots + {}^{2n}C_n$

$[{}^{n}C_r + {}^{n}C_{r+1} = {}^{n+1}C_{r+1}]$

$= ({}^{(n+3)}C_{n+1} + {}^{n+3}C_n) + \ldots + {}^{2n}C_n$

$= {}^{2n+1}C_{n+1}$

**(M 2)**

$(t+1)^n + (t+1)^{n+1} + (t+1)^{n+2} + \ldots + (t+1)^{2n} = S$(say)

To find coefficient of $t^n$ from above series.

$A = (t+1)^n \qquad R = (t+1)$

$$S = A\left[\frac{R^n - 1}{R - 1}\right] = (x+1)^n \left[\frac{(x+1)^{n+1} - 1}{(x+1) - 1}\right]$$

$$S = \underset{\vee}{\frac{(x+1)^{2n+1} - (x+1)^n}{}} = \underset{\vee}{\frac{(x+1)^{2n+1}}{}} - \underset{\vee}{\frac{(x+1)^n}{}}$$

$\therefore$ coefficient of $t^n$ in $S = {}^{2n+1}C_{n+1}$ (coefficient of $t^{n+1}$ in $(1+t)^{2n+1}$)

$\therefore\ b_n = {}^{2n+1}C_{n+1}$

### Term Independent of x

**Step-I:** Write general term of the expansion.
**Step-II:** Equate the exponent of x to zero

**Q.** **Find the term independent of x in** $\left(2x^2 - \frac{1}{3x}\right)^6$

**Sol.** $T_{r+1} = {}^6C_r \left(2x^2\right)^{6-r} \left(-\frac{1}{3x}\right)^r$

$= {}^6C_r 2^{6-r} x^{12-2r} \left(-\frac{1}{3}\right)^r \cdot \frac{1}{x^r}$

$= {}^6C_r 2^{6-r} \left(-\frac{1}{3}\right)^r x^{12-3r}$

For term independent of x, $12 - 3r = 0$

$r = 4$

Hence $T_{4+1} = T_5$ or $5^{th}$ term is independent of x

$T_5 = {}^6C_4 2^{6-4} \left(-\frac{1}{3}\right)^4 x^0 = 15 \times 4 \times \frac{1}{81} = \frac{20}{27}$

**Q.** **If the term independent of x in the expansion of** $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$ **is k, then 18k is equal to**

**(A) 5** **(B) 9** **(C) 7** **(D) 11**

**Sol.** **(C)**

$T_{r+1} = {}^9C_r \left(\frac{3}{2}x^2\right)^{9-r} \left(\frac{-1}{3x}\right)^r$

$= {}^9C_r \left(\frac{3}{2}\right)^{9-r} x^{18-2r} \left(-\frac{1}{3}\right)^r \left(\frac{1}{x}\right)^r$

For term independent of x, $18 - 3r = 0 \Rightarrow r = 6$

So, term independent of $x = {}^9C_6 \left(\frac{3}{2}\right)^3 \left(\frac{-1}{3}\right)^6 = \frac{7}{18}$

$\therefore\ 18K = 18 \times \frac{7}{18} = 7$

**Q.** **Let $\alpha > 0, \beta > 0$ be such that $\alpha^3 + \beta^2 = 4$. If the maximum value of the term independent of x in the binomial expansion of $\left(\alpha x^{\frac{1}{9}} + \beta x^{\frac{-1}{6}}\right)^{10}$ is 10k, then k is equal to**

**(A) 176** **(B) 336** **(C) 352** **(D) 84**

**Sol.** **(B)**

$$T_{r+1} = {}^{10}C_r \left(\alpha x^{\frac{1}{9}}\right)^{10-r} \left(\beta x^{\frac{-1}{6}}\right)^r$$

For term independent of x, $\frac{10-r}{9} - \frac{r}{6} = 0 \Rightarrow r = 4$

$\therefore T_5 = {}^{10}C_4\, \alpha^6\beta^4$

$$\frac{\alpha^3 + \beta^2}{2} \geq \left(\alpha^3\beta^2\right)^{\frac{1}{2}}$$

$$\left(\frac{4}{2}\right)^2 \geq \alpha^3\beta^2 \Rightarrow (\alpha^3\beta^2)_{max} = 4$$

$\therefore (T_5)_{max} = 10\ k$

${}^{10}C_4(4)^2 = 10\ k$

$k = 336$

### Middle Term

The middle term in the expansion of $(x + y)^n$ depends upon the value of n.

#### (i) If n is even

Total number of terms in the expansion = n + 1 (odd)

There is only one middle term

$\left(\frac{n}{2} + 1\right)^{th}$ term is the middle term

#### (ii) If n is odd

Total number of terms in the expansion = n + 1 (even)

There is two middle term

$\left(\frac{n+1}{2}\right)^{th}$ term and $\left(\frac{n+1}{2} + 1\right)^{th}$ term are two middle term.

**Q.** **Find middle terms in the expansion of** $\left(2x^2 - \frac{1}{3x}\right)^{11}$

**Sol.** $n = 11$

$\frac{n+1}{2} = 6$

$\frac{n+1}{2} + 1 = 7$

$T_6$ and $T_7$ are middle terms

$T_6 = -{}^{11}C_5 \left(2x^2\right)^6 \left(\frac{1}{3x}\right)^5$ ;

$T_7 = {}^{11}C_6 \left(2x^2\right)^5 \left(-\frac{1}{3x}\right)^6 = -\frac{9856}{81}x^7$

$= \frac{4928}{243}x^4$

**Note :**

**(i)** Middle term has greatest binomial coefficient and if there are two middle terms their binomial coefficients will be equal

**(ii)** ${}^nC_r$ will be maximum

When $r = \frac{n}{2}$ if $n \in$ even

When $r = \frac{n-1}{2}$ or $\frac{n+1}{2}$ if $n \in$ odd

**Q.** **In the expansion of $(1 + x)^{10}$, coefficients of $(4r + 5)^{th}$ term = coefficient of $(2r + 1)^{th}$ term. Find r**

**Sol.** ${}^{10}C_{4r+4} = {}^{10}C_{2r}$

$4r + 4 = 2r$ or $10 = (4r + 4) + 2r \Rightarrow r = 1$

$2r = -4$ (not possible)

$[{}^nC_x = {}^nC_y \Rightarrow x + y = n$ or $x = y]$

**Q.** **Greatest value of term independent of x in** $\left(x\sin\alpha + \frac{\cos\alpha}{x}\right)^{10}$ $x \in R$

**Sol.** $T_{r+1} = {}^{10}C_r (x\sin\alpha)^{10-r} \left(\frac{\cos\alpha}{x}\right)^r$

For term independent of x, $10 - 2r = 0 \Rightarrow r = 5$

$T_6 = {}^{10}C_5(\sin\alpha)^5 (\cos\alpha)^5$

$= \frac{{}^{10}C_5}{2^5} \times (2\sin\alpha\sin\alpha)^5 = \frac{{}^{10}C_5}{2^5}(\sin 2\alpha)^5$

$(T_6)_{max} = \frac{{}^{10}C_5}{2^5}$

**Q.** **In the expansion of** $\left(\sqrt{2} + \sqrt{7}\right)^{15}$**, number of rational terms are:**

**Sol.** $T_{r+1} = {}^{15}C_r \left(\sqrt{2}\right)^{15-r} \left(\sqrt{7}\right)^r$

For rational terms 15 – r and r should be even which is not possible

Hence, all terms will be irrational
So, number of rational terms is zero.

**Q.** $\left(4^{\frac{1}{3}} + \frac{1}{6^{\frac{1}{4}}}\right)^{20}$ **. Find number of rational terms.**

**Sol.** $T_{r+1} = {}^{20}C_r \left(4^{\frac{1}{3}}\right)^{20-r} \left(6^{\frac{-1}{4}}\right)^r$

For rational terms, 'r' should be multiple of 4

$\therefore$ Possible values of r is 0, 4, 8, 12, 16, 20

but at r = 8 and 20 only $\frac{20-r}{3}$ is integer

$\therefore$ r = 8,20

Hence, number of rational terms are 2

## Multinomial theorem

In the expansion of $(x_1 + x_2 + x_3 + x_4)^n$ each term has different power and their power is always n.

$$(x_1 + x_2 + x_3 + x_4 + \ldots)^n = \sum \frac{n!}{a.b.c....} x_1^a x_2^b x_3^c$$

where $a + b + c + \ldots = n$

$0 \le a, b, c, \ldots \le n$

Total number of terms in the expansion is equal to number of ways in which different power can be distributed to $x_1, x_2, x_3 \ldots x_n$ and sum of their powers is always n. Number of terms in expansion of $(x_1 + x_2 + x_3 + \ldots x_n)^r$ is

$= {}^{n+r-1}C_{r-1}$

**Q.** **Find number of distinct terms in the expansion of**
**(a) $(x + y + z)^2$** **(b) $(x + y + z)^8$**

**Sol.** **(a)** $n = 2, \ r = 3$

$${}^{n+r-1}C_{r-1} = {}^{3+2-1}C_{3-1} = {}^4C_2 = 6$$

**(b)** $n = 8 \quad r = 3$

$${}^{n+r-1}C_{r-1} = {}^{8+3-1}C_{3-1} = {}^{10}C_2 = 45$$

## Greatest Binomial Coefficient

$(1 + x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \ldots {}^nC_n x^n$

Middle Binomial coefficient will be greatest i.e..

⇒ If n is even, number of binomial coefficient will be odd then their will be one middle binomial coefficient which will be greatest.

i.e ${}^nC_{n/2}$ will be greatest coefficient.

⇒ If n is odd

⇒ ${}^nC_{\frac{n+1}{2}} = {}^nC_{\frac{n-1}{2}}$ will be greatest coefficient.

**Points to Remember!!!**

**We already know that**

$$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$$

## Greatest term of expansion

As we know

$$(1 + x)^n = {}^nC_0 + {}^nC_1 + {}^nC_2 x^2 + {}^nC_3 x^3 + \ldots\ldots\ldots\ldots + {}^nC_n x^n$$

$$\vdots \quad \vdots \quad \vdots \qquad \vdots$$

$$\vdots \quad \vdots \quad \vdots \qquad \vdots$$

$$(1+x)^n = T_1 + T_2 + T_3 + \ldots + T_r + T_{r+1} + \ldots + T_{n+1}$$

To get greatest term find r until

$\Rightarrow \quad |T_{r+1}| \geq |T_r|$

$\Rightarrow \quad |{}^nC_r\, x^r| \geq |{}^nC_{r-1}\, x^{r-1}|$

$= {}^nC_r\, |x^r| \geq {}^nc_{r-1}\, |x^{r-1}|$ (as ${}^nc_r$ is always +ve)

$$\frac{n!}{(n-r)!\,r!}\,|x| \geq \frac{n!}{(r-1)!\,(n-r+1)!}$$

$$= \frac{|x|}{r} \geq \frac{1}{n-r+1}$$

$$= n|x| - r|x| + |x| \geq r$$

$$|x|\,(n+1) \geq r\,(1+|x|)$$

$$r \leq \frac{(n+1)\,|x|}{1+|x|} = m \quad \text{(Lets say)}$$

**Case I :** If $m \notin I$, then

$T_{[m]+1}$ will be greatest term in magnitude

$\Rightarrow |T_{max}| = [T_{[m]+1}] = {}^nC_m\, x^m$

**Case II :** If $m \in I$, then $T_{[m]}$, $T_{[m]+1}$ will be equal in magnitude

$|T_{max}| = |T_{[m]}| = |T_{[m]+1}|$

**Note :** The greatest coefficient in binomial is equal to greatest term when $x = 1$.

**Q.** **$^{57}C_r$ find maximum and minimum binomial coefficient.**

**Sol.** $^{57}C_{57} = {}^{57}C_0$ is minimum coefficient.

$^{57}C_{29} = {}^{57}C_{28}$ is maximum binomial.

**Q.** **Find greatest term in the expansion of $(4+5x)^{38}$ where $x = \frac{1}{2}$.**

**Sol.** $$(4+5x)^{38} = 4^{38}\left(1+\frac{5}{4}x\right)^{38}$$

$$\Rightarrow r \leq \frac{(39)\cdot\left(\frac{5}{4}\times\frac{1}{2}\right)}{\left(1+\frac{5}{8}\right)} \Rightarrow r \leq \frac{39\times\frac{5}{8}}{\frac{13}{8}}$$

$$\Rightarrow \quad r \leq 15$$

so $|T_r|_{max} = |T_{15}| = |T_{16}|$
As all terms are positive, hence
$T_{max} = |T_{max}| = |T_{15}| = T_{15} = T_{16}$.

$$4^{38} \cdot {}^{38}C_{14}\left(\frac{5}{8}\right)^{19} = 4^{38}\,{}^{38}C_{15}\left(\frac{5}{8}\right)^{15}$$

**Q.** **In the expansion of $y = 1 + (1 + x) + (1 + x)^2 + \ldots + (1 + x)^{19}$, if the coefficient of $x^p$ is greatest. Find p**

**Sol.** $$y = \frac{(1+x)^{20} - 1}{(1+x) - 1} = \frac{(1+x)^{20} - 1}{x}$$

$\therefore$ coefficient of $x^p$ in y is ${}^{20}C_{p+1}$
${}^{20}C_{p+1}$ is greatest when $p + 1 = 10$
$\Rightarrow p = 9$.

**Q.** **If $x = \frac{1}{3}$, find the greatest term in the expansion of $(1 + 4x)^8$**

**Sol.** Let $T_{r+1}$ be greatest

$T_{r+1} \geq T_r$

${}^8C_r(4x)^r \geq {}^8C_{r-1}(4x)^{r-1}$

$$\frac{8!}{r!(8-r)!}(4x) \geq \frac{8!}{(r-1)!(9-r)!}$$

$$\frac{4}{3}\frac{1}{r} \geq \frac{1}{9-r} \Rightarrow 36 - 4r \geq 3r$$

$$\therefore r \leq \frac{36}{7} = 5.1 \text{ (approx)}$$

| | | | |
|---|---|---|---|
| at r = 5 | $T_6 \geq T_5$ | Also, | $T_7 \leq T_6$ |
| at r = 4 | $T_5 \geq T_4$ | | $T_8 \leq T_7$ |
| at r = 3 | $T_4 \geq T_3$ | | $T_9 \leq T_8$ |
| at r = 2 | $T_3 \geq T_2$ | | |
| at r = 1 | $T_2 \geq T_1$ | | |

Hence, greatest is $T_6$

$$T_6 = T_{5+1} = {}^8C_5\left[4\left(\frac{1}{3}\right)\right]^5 = \frac{8\times7\times6}{6}\times\frac{4^5}{3^5} = \frac{57344}{243}$$

**Q.** **Find which term is numerically greatest term in $(3-2x)^9$ at $x = 1$**

**Sol.** $|T_{r+1}| \geq |T_r|$

$|{}^9C_r 3^{9-r}(-2x)^r| \geq |{}^9C_{r-1}\, 3^{10-r}(-2x)^{r-1}|$

$$\left|\frac{9!}{r!(9-r)!}(2)\right| \geq \left|3\frac{9!}{(10-r)!(r-1)!}\right|$$

Solving $r \leq 4$

$\therefore$ at $r = 4$ $\quad T_5 \geq T_4$ $\quad$ Also, $\quad T_6 \leq T_5$

at $r = 3$ $\quad T_4 \geq T_3$ $\quad T_7 \leq T_6$

at $r = 2$ $\quad T_3 \geq T_2$ $\quad \vdots$

at $r = 1$ $\quad T_2 \geq T_1$ $\quad T_{10} \leq T_9$

$\therefore$ $T_5$ is numerically greatest term.

**Q.** **Given $T_4$ in the expansion of $\left(2+\frac{3x}{8}\right)^{10}$ has maximum numerical value. Find range of x.**

**Sol.** $|T_4| > |T_5|$ and $|T_4| > |T_3|$

$$\left|{}^{10}C_3 2^7\left(\frac{3x}{8}\right)^3\right| > \left|{}^{10}C_4 2^6\left(\frac{3x}{8}\right)^4\right| \;\&\; \left|{}^{10}C_3 2^7\left(\frac{3x}{8}\right)^3\right| > \left|{}^{10}C_2 2^8\left(\frac{3x}{8}\right)^2\right|$$

$$\left|\frac{1}{4}\left(\frac{3x}{8}\right)\right| < \left|\frac{2}{7}\right| \qquad \left|\frac{3x}{8}\right|\times\frac{1}{3} > \frac{2}{8}$$

$$|x| < \frac{64}{21} \qquad |x| > 2$$

$$x \in \left(-\frac{64}{21}, -2\right) \cup \left(2, \frac{64}{21}\right)$$

## More Questions on Binomial Expansion

**Q.** **If {P} denotes the fractional part of the number P, then $\left\{\frac{3^{200}}{8}\right\}$, is equal to**

**(A)** $\frac{5}{8}$ **(B)** $\frac{1}{8}$ **(C)** $\frac{7}{8}$ **(D)** $\frac{3}{8}$

**Sol.** **(B)**

$3^{200} = (1 + 8)^{100} = 1 + {}^{100}C_1\, 8^1 + {}^{100}C_2\, 8^2 + \ldots$

$= 1 + 8k,\ k \in \text{integer}$

$\therefore \quad \left\{\frac{3^{200}}{8}\right\} = \left\{\frac{1+8k}{8}\right\} = \left\{\frac{1}{8} + k\right\} = \left\{\frac{1}{8}\right\} = \frac{1}{8}$

**Q.** **The remainder when $(15^{23} + 23^{23})$ is divided by 19 is**

**(A) 4** **(B) 15** **(C) 0** **(D) 18**

**Sol.** **(C)**

$(19 - 4)^{23} + (19 + 4)^{23}$

$= 2[{}^{23}C_0 19^{23} + {}^{23}C_2\, 19^{21}\, 4^2 + \ldots + {}^{23}C_{22}\, 19^1 4^{22}]$

= a multiple of 19.

Hence, remainder = 0

**Q.** **Prove that $({}^{72}C_{36} - 1)$ is divisible by 73.**

**Sol.** ${}^{72}C_{36} = {}^{73}C_{36} - {}^{72}C_{35}\ ({}^nC_{r+1} = {}^{n+1}C_{r+1} - {}^nC_r)$

$= {}^{73}C_{36} - ({}^{73}C_{35} - {}^{72}C_{34})$

$= {}^{73}C_{36} - {}^{73}C_{35} + ({}^{73}C_{34} - {}^{72}C_{33})$

${}^{72}C_{36} - 1 = {}^{73}C_{36} - {}^{73}C_{35} + {}^{73}C_{34} - {}^{73}C_{33} + {}^{73}C_{32} - \ldots - {}^{73}C_1 + {}^{72}C_0 - 1]$

$= [{}^{73}C_{36} - {}^{73}C_{35} + {}^{73}C_{34} - {}^{73}C_{33} + {}^{73}C_{32} - \ldots - {}^{73}C_1]$

Each of the term contains a factor 73.

Hence, $({}^{72}C_{36} - 1)$ is divisible by 73.

## Questions on Expansion of $(x + y)^n \pm (x-y)^n$

$(x + y)^n = {}^nC_0x^n + {}^nC_1 x^{n-1}y^1 + {}^nC_2 x^{n-2} y^2 + \ldots$ ... (i)

$(x-y)^n = {}^nC_0 x^n - {}^nC_1 x^{n-1} y + {}^nC_2 x^{n-2}y^2 + \ldots$ ...(ii)

Adding (i) and (ii)

$(x+y)^n + (x-y)^n = 2\,[{}^nc_0 x^n + {}^nc_2 x^{n-2} y^2 + {}^nc_4 x^{n-4} y^4 + \ldots$

$(x + y)^n - (x-y)^n = 2\,[{}^nc_1 x^{n-1}y + {}^nc_3 x^{n-3} y^3 + {}^nc_5 x^{n-5} y^5 + \ldots$

**Q.** **Value of $\left(1+\sqrt{2}\right)^7 + \left(1-\sqrt{2}\right)^7$**

**(A) 402** **(B) 458** **(C) 478** **(D) 498**

**Sol.** **(C)**

$$\left(1+\sqrt{2}\right)^7 = 1 + 7C_1\left(\sqrt{2}\right)^1 + 7C_2\left(\sqrt{2}\right)^2 + 7C_3\left(\sqrt{2}\right)^3 + \ldots +$$

$$\left(1-\sqrt{2}\right)^7 = 1 - 7C_1\left(\sqrt{2}\right)^1 + 7C_2\left(\sqrt{2}\right)^2 - 7C_3\left(\sqrt{2}\right)^3 + \ldots$$

---

$$\left(1+\sqrt{2}\right)^7 + \left(1-\sqrt{2}\right)^7 = 2\left[1 + {}^7C_2\,(2) + {}^7C_4 2^2 + {}^7C_6 2^3\right]$$

$= 2[1 + 42 + 140 + 56]$

$= 478$

**Q.** **Find value of $\left(\sqrt{3}+3\right)^5 - \left(\sqrt{3}-3\right)^5$**

**Sol.** $\sqrt{3}^5\left[\left(1+\sqrt{3}\right)^5 - \left(1-\sqrt{3}\right)^5\right]$

$$= \left(\sqrt{3}\right)^5\left[2\left\{{}^5C_1\left(\sqrt{3}\right)^1 + {}^5C_3\left(\sqrt{3}\right)^3 + {}^5C_5\left(\sqrt{3}\right)^5\right\}\right]$$

$= 54[5 + 30 + 9]$

$= 54 \times 44 = 2376$

**Q.** **Which is larger: $(99^{50} + 100^{50})$ or $(101)^{50}$**

**Sol.** $99^{50} + 100^{50} \leftrightarrow (101)^{50}$

$100^{50} \leftrightarrow (1 + 100)^{50} - (1 - 100)^{50}$

$100^{50} \leftrightarrow (C_0 + C_1x + C_2x^2 + \ldots) - (C_0 - C_1x + C_2x^2 \ldots)$ $(x = 100)$

$100^{50} \leftrightarrow 2[{}^{50}C_1\,100 + {}^{50}C_3 100^3 + \ldots + {}^{50}C_{49} 100^{49}]$

$100^{50} \leftrightarrow 2[{}^{50}C_1 100 + {}^{50}C_3 100^3 + \ldots] + 100^{50}$

Hence right hand side is bigger than left hand side
$\Rightarrow 99^{50} + 100^{50} < (101)^{50}$

**Q.** **If $\alpha$ and $\beta$ be the coefficients of $x^4$ and $x^2$ respectively in the expansion of $(x+\sqrt{x^2-1})^6 + (x-\sqrt{x^2-1})^6$ then:**

**(A) $\alpha + \beta = 0$** **(B) $\alpha - \beta = -132$** **(C) $\alpha + \beta = 60$** **(D) $\alpha - \beta = 60$**

**Sol.** **(B)**

$(x+y)^6 + (x-y)^6 = 2\{{}^6C_0x^6 + {}^6C_2x^4y^2 + {}^6C_4x^2y^4 + {}^6C_6y^6\}$

$\therefore (x+\sqrt{x^2-1})^6 + (x-\sqrt{x^2-1})^6$

$2\{x^6 + {}^6C_2x^4(x^2-1) + {}^6C_4x^2(x^2-1)^2 + {}^6C_6(x^2-1)^3\}$

$\therefore \alpha = 2[-{}^6C_2 - 2\,{}^6C_4 - 3] = -96$

$\beta = 2[{}^6C_4 + 3] = 36$

$\therefore \alpha - \beta = -96 - 36 = -132$

**Q.** **Find the integral part of $(2+\sqrt{3})^6$**

**Sol.** Let $(2+\sqrt{3})^6 = I + f,\ f > 0$

$+ \&\ (2-\sqrt{3})^6 = f_1,\ 0 < f_1 < 1$

$(2+\sqrt{3})^6 + (2-\sqrt{3})^6 = I + f + f_1$

$2[{}^6C_0 2^6 + {}^6C_2 2^4 3^1 + {}^6C_4 2^2 3^2 + {}^6C_6 3^3] = I + (f + f_1)$

$2[64 + 15 \times 48 + 15 \times 36 + 27] = I + (f + f_1)$

Even integer $= I + (f + f_1)$

$f_1 + f =$ integer (odd) ...(i)

Now,

$0 < f < 1$

$0 < f_1 < 1$

$0 < f + f_1 < 2$ ...(ii)

from (i) and (ii)

$f + f_1 = 1$

$\therefore\ 2[64 + 720 + 540 + 27] = I + 1$

$I = 2701$

**Point to Remember!!!**

$R = I + F$

where
R = Real numbers,
I = Integral part,
f = Fractional part where $f \in [0, 1)$

**Q.** **Find the value of $(2+\sqrt{3})^6(1-f)$, where f is fraction part of $(2+\sqrt{3})^6$**

**Sol.** From last question: $1 - f = f_1 = (2-\sqrt{3})^6$

$\therefore\ (2+\sqrt{3})^6(2-\sqrt{3})^6 = (4-3)^6 = 1$

**Q.** **Prove that integer part of $(3+\sqrt{7})^n$ is odd integer**

**Sol.**
$$\left.\begin{array}{l}(3+\sqrt{7})^n = I + f \quad , \quad 0 < f < 1 \\ +(3-\sqrt{7})^n = f_1 \quad , \quad 0 < f_1 < 1\end{array}\right\} \Rightarrow 0 < f_1 + f < 2 \qquad \ldots(i)$$

---

$2[{}^nC_0 3^n + {}^nC_2 3^{n-2}7^1 + {}^nC_4\ 3^{n-4}7^2 + \ldots] = I + f + f_1$

Even integer $= I + f + f_1$

$\Rightarrow f + f_1 =$ integer $\qquad \ldots(ii)$

from (i) & (ii) $f + f_1 = 1 \qquad \ldots(iii)$

$\therefore$ even integer $- (f + f_1) = I$

even integer $- 1 = I$

$\therefore$ I is an odd integer

**Q.** **Show that integral part of $(5+3\sqrt{3})^{2n+1}$ is even.**

**Sol.** $(5+3\sqrt{3})^{2n+1} = I + f$

$+\ (5-3\sqrt{3})^{2n+1} = -f_1 \qquad f_1 \in (0, 1)$

$2[{}^{2n+1}C_0\ 5^{2n+1} + {}^{2n+1}C_2\ 5^{2n-1}(3\sqrt{3})^2 + \ldots] = I + f - f_1$

Even integer $= I + f - f_1$

$\therefore\ f - f_1$ integer between $-1$ & $1 \Rightarrow f - f_1 = 0$

$$\left(\begin{array}{c} 0 < f < 1 \\ \hline -1 < -f_1 < 0 \\ \hline -1 < f - f_1 < 1 \end{array}\right)$$

$\therefore$ even integer $= I + 0$

$\therefore$ I = Even integer.

**Q.** **Let I denotes the integral part & f the proper fraction part of $(3+\sqrt{5})^n$ where $n \in N$ and if $\rho$ denotes the rational part and $\sigma$ the irrational part of the same, show that $\rho = \frac{1}{2}(I+1)$ and $\sigma = \frac{1}{2}(I + 2f - 1)$**

**Sol.** $(3+\sqrt{5})^n = I + f = {}^nC_0 3^n + {}^nC_1 3^{n-1}(\sqrt{5})^1 + \ldots$

$(3-\sqrt{5})^n = f_1 = {}^nC_0 3^n - {}^nC_1 3^{n-1}(\sqrt{5})^1 + \ldots$

$\therefore \quad I + f = \rho + \sigma$ ...(i)

$f_1 = \rho - \sigma$ ...(ii)

---

(i) + (ii) given,

$I + f + f_1 = 2\rho$

$I + 1 = 2\rho \qquad (\because f + f_1 = 1)$

$\rho = \frac{I+1}{2}$

(i) – (ii) gives,

$I + f - f_1 = 2\sigma$

$\frac{I + f - (1-f)}{2} = \sigma$

$\therefore \quad \sigma = \frac{I + 2f - 1}{2}$

## Sum of coefficients of Binomial

### *Sum of all binomial coefficients in expansion of $(x+y)^n$*

**$^nC_0 + {}^nC_1 + {}^nC_2 + \ldots + {}^nC_n = 2^n$** ...(i)

**Proof :**

$(1+x)^n = {}^nC_0 x^0 + {}^nC_1 x^1 + {}^nC_2 x^2 + \ldots + {}^nC_n x^n$

Put $x = 1$

$(1+1)^n = {}^nC_0 + {}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots {}^nC_n = 2^n$

**$^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \ldots = 0$** ...(ii)

**Proof :**

$(1-x)^n = {}^nC_0 x^0 - {}^nC_1 x + {}^nC_2 x^2 - {}^nC_3 x^3 + \ldots$

Put $x = 1$

$0 = {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \ldots$

**$^nC_0 + {}^nC_2 + {}^nC_4 + \ldots = 2^{n-1}$**

**$^nC_1 + {}^nC_3 + {}^nC_5 + \ldots = 2^{n-1}$**

**Proof :**

(i) + (ii)

$2\,{}^nC_0 + 2\,{}^nC_2 + 2\,{}^nC_4 + \ldots = 2^n$

${}^nC_0 + {}^nC_2 + {}^nC_4 + \ldots = 2^{n-1}$

Similarly (i) – (ii) gives

${}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots = 2^{n-1}$

**Note :**

$(x + 2y)^2 = {}^2C_0x^{2-0}(2y)^0 + {}^2C_1x^{2-1}(2y)^1 + {}^2C_2x^{2-2}(2y)^2$

$= 1x^2 + 4xy + 4y^2$

Here, ${}^2C_0$, ${}^2C_1$, ${}^2C_2$ are binomial coefficients and 1, 4, 4 are coefficients.

To find sum of binomial coefficients in expansion of $(x + 2y)^2$, use formula (1) and to find sum of all coefficients in the expansion of $(x+2y)^2$ put $x = y = 1$, i.e. $[1 + 2(1)]^2 = 9$

**Q.** **Sum of all coefficient in the expansion of $\left(x+\frac{1}{x}\right)^n$ is 4096. Find greatest binomial coefficients.**

**Sol.** Put $x = 1$

$(1 + 1)^n = 4096 = 2^{12} \Rightarrow n = 12$

Greatest binomial coefficient $= {}^{12}C_6$

**Q.** **Sum of last 10 coefficients in the expansion of $(1 + x)^{19}$**

**Sol.**

$$\begin{array}{ll} & S = {}^{19}C_{19} + {}^{19}C_{18} + \ldots + {}^{19}C_{10} \\ + & S = {}^{19}C_0 + {}^{19}C_1 + \ldots + {}^{19}C_9 \qquad ({}^nC_r = {}^nC_{n-r}) \\ \hline & 2S = 2^{19} \Rightarrow S = 2^{18} \end{array}$$

**Q.** **Sum of last 9 coefficients in the expansion of $(1 + x)^{18}$**

**Sol.**

$$\begin{array}{ll} & S = {}^{18}C_{18} + {}^{18}C_{17} + \ldots + {}^{18}C_{10} \\ + & S = {}^{18}C_0 + {}^{18}C_1 + \ldots + {}^{18}C_8 \\ \hline \end{array}$$

$2S = ({}^{18}C_0 + {}^{18}C_1 + \ldots + {}^{18}C_8 + {}^{18}C_9 + {}^{18}C_{10} + \ldots + {}^{18}C_{18}) - {}^{18}C_9$

$S = \dfrac{2^{18} - {}^{18}C_9}{2}$

**Q.** **Sum of all coefficients in $(1 + x - 3x^2)^{2163}$**

**Sol.** Put $x = 1$

$(1 + 1 - 3)^{2163} = -1$

**Q.** **The sum of the coefficients in the expansion of $(a + 2b + c)^{10}$ is**

**(A)** $4^{10}$ **(B)** $3^{10}$ **(C)** $2^{10}$ **(D)** $10^4$

**Sol.** **(A)**

Put $a = b = c = 1$

$(1 + 2(1) + 1)^{10} = 4^{10}$

**Q.** **Find the sum of all coefficients in the expansion of**

**(a)** $(x - y + z)^2$ **(b)** $(2x^2 - 3x + 3)^{2021}$

**(c)** $(x + 3)^{2020}$ **(d)** $(x - 1)^{99}$

**Sol.**

**(a)** put $x = y = z = 1$

$(1 - 1 + 1)^2 = 1$

**(b)** put $x = 1$, $[2(1)^2 - 3(1) + 3]^{2021} = 2^{2021}$

**(c)** put $x = 1$, $(1 + 3)^{2020} = 4^{2020}$

**(d)** put $x = 1$, $(1 - 1)^{99} = 0$

**Q.** **The sum of coefficients of integral powers of x in the binomial expansion of $\left(1 - 2\sqrt{x}\right)^{50}$ is**

**(A)** $\frac{1}{2}\left(2^{50} + 1\right)$ **(B)** $\frac{1}{2}\left(3^{50} + 1\right)$ **(C)** $\frac{1}{2}\left(3^{50}\right)$ **(D)** $\frac{1}{2}\left(3^{50} - 1\right)$

**Sol.** **(B)**

$$\left(1 - 2\sqrt{x}\right)^{50} = {}^{50}C_0 - {}^{50}C_1\left(2\sqrt{x}\right)^1 + {}^{50}C_2\left(2\sqrt{x}\right)^2 - {}^{50}C_3\left(2\sqrt{x}\right)^3$$

$$+ {}^{50}C_4\left(2\sqrt{x}\right)^4 - {}^{50}C_5\left(2\sqrt{x}\right)^5 + \ldots + (-1)^{50}\,{}^{50}C_{50}\left(2\sqrt{x}\right)^{50} \quad \ldots(i)$$

Required sum $= {}^{50}C_0 + {}^{50}C_2\,2^2 + {}^{50}C_4 2^4 + \ldots + {}^{50}C_{50}2^{50}$

$$(1 + 2\sqrt{x})^{50} = {}^{50}C_0 + {}^{50}C_1(2\sqrt{x})^1 + {}^{50}C_2(2\sqrt{x})^2 + {}^{50}C_3(2\sqrt{x})^3 + \ldots + {}^{50}C_{50}(2\sqrt{x})^{50} \quad \ldots(ii)$$

(i) + (ii) gives

$$\left(1-2\sqrt{x}\right)^{50}+\left(1+2\sqrt{x}\right)^{50}=2\left[{}^{50}C_0+{}^{50}C_2\left(2\sqrt{x}\right)^2+{}^{50}C_4\left(2\sqrt{x}\right)^4+\ldots+{}^{50}C_{50}\left(2\sqrt{x}\right)^{50}\right]$$

put x = 1 both side

$$\frac{(1-2)^{50}+(1+2)^{50}}{2}={}^{50}C_0+{}^{50}C_2 2^2+{}^{50}C_4 2^4+\ldots+{}^{50}C_{50}2^{50}$$

On solving L.H.S. is $\frac{1}{2}\left(3^{50}+1\right)$.

**Q.** **$(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n}$**
**(i) Show that $a_0 + a_1 + a_2 + \ldots + a_{2n} = 3n$**

**Sol.** **(M 1)**

$[1+(x+x^2)]^n = {}^nC_0 + {}^nC_1(x+x^2) + {}^nC_2(x+x^2)^2 + \ldots$

Put x = 1

$(1+1+1)^n = a_0 + a_1 + \ldots + a_{2n}$

$\therefore\ a_0 + a_1 + \ldots + a_{2n} = 3^n$

**(M 2)**

$[x^2+(x+1)]^n = {}^nC_0(x^2)^0(x+1)^n + {}^nC_1(x^2)^1(x+1)^{n-1} + {}^nC_2(x^2)^2(x+1)^{n-2} + \ldots$

put x = 1

$(1+1+1)^n = a_0 + a_1 + a_2 + \ldots$

$\therefore\ a_0 + a_1 + a_2 + \ldots = 3^n$

**(ii) $a_0 - a_1 + a_2 - a_3 + \ldots + a_{2n} = 1$**

**Sol.** Put x = – 1 in given expression

$[1-1+(-1)^2]^n = a_0 - a_1 + a_2 + \ldots + a_{2n}$

$\therefore\ a_0 - a_1 + a_2 - a_3 + \ldots = 1$

**(iii) $a_1 + a_3 + a_5 + \ldots + a_{2n-1} =$**

**Sol.** Part (i) – (ii) gives

$2(a_1 + a_3 + \ldots) = 3^n - 1$

$\therefore\ a_1 + a_3 + a_5 + \ldots = \frac{3^n - 1}{2}$

**(iv)** $a_0 + a_2 + a_4 + \ldots + a_{2n} = \dfrac{3^n + 1}{2}$

**Sol.** Part (i) + (ii) gives

$2[a_0 + a_2 + \ldots] = 3^n + 1$

$$a_0 + a_2 + a_4 + \ldots = \frac{3^n + 1}{2}$$

**Important facts and properties to find sum of series.**

**(i)** $r \cdot {}^nC_r = n \cdot {}^{n-1}C_{r-1}$

$$\text{L.H.S.} = r \cdot \frac{n!}{r!(n-r)!} = \frac{n(n-1)!}{(n-r)!(r-1)!} = n \cdot {}^{n-1}c_{r-1}$$

**(ii)** $\dfrac{{}^nc_r}{n} = \dfrac{{}^{n-1}c_{r-1}}{r}$ (using above property)

Replace $n \to n + 1$

$r \to r + 1$

$$\frac{{}^{n+1}c_{r+1}}{n+1} = \frac{{}^nc_r}{r+1}$$

**(iii)** ${}^kc_k + {}^{k+1}c_k + {}^{k+2}c_k + {}^{k+3}c_k + \ldots + {}^nc_k = {}^{n+1}c_{k+1}$

**Proof :**

We know

${}^kc_k + {}^{k+1}c_k = {}^{k+2}c_{k+1}$

Similarly

${}^{k+2}c_{k+1} + {}^{k+2}c_k = {}^{k+3}c_{k+1}$

${}^{k+3}c_{k+1} + {}^{k+3}c_k = {}^{k+4}c_{k+1}$

$\vdots \qquad \vdots$

Same logic we get sum = ${}^{n+1}c_{c+1}$

**(iv)** $\displaystyle\sum_{r=0}^{n} r \cdot {}^nc_r = n \cdot 2^{n-1}$

**Proof :** $\displaystyle\sum_{r=0}^{n} r \cdot {}^nc_r = \sum_{r=1}^{n} r \cdot {}^nc_r = 1 \cdot {}^nC_1 + 2 \cdot {}^nC_2 + 3 \cdot {}^nC_3 + \ldots$

$$\sum_{r=1}^{n} n \cdot {}^{n-1}c_{r-1} = n \sum_{r=1}^{n} {}^{n-1}c_{r-1} = n \cdot 2^{n-1}$$

So, $\sum_{r=1}^{n} r \cdot {}^{n}c_r = n \cdot 2^{n-1}$

**(v)** $\sum_{r=0}^{n} r^2 \cdot {}^{n}c_r = 2^{n-2}(n^2 + n)$

**Proof :** $\sum_{r=1}^{n} r^2 \cdot {}^{n}c_r$ (As for r = 0, term → 0)

$= \sum_{r=1}^{n} r \cdot n \, {}^{n-1}c_{r-1} = n\sum_{r=1}^{n} (r+1-1) \, {}^{n-1}c_{r-1}$

$= n\sum_{r=1}^{n} (r-1) \, {}^{n-1}c_{r-1} + n\sum_{r=1}^{n} {}^{n-1}c_{r-1}$

$= n\sum_{r=2}^{n} (r-1) \, {}^{n-1}c_{n-1} + \left(n \cdot 2^{n-1}\right)$

$= n\sum_{r=2}^{n} (n-1) \, {}^{n-2}c_{r-2} + n \cdot 2^{n-1}$

$= (n-1)\, n \cdot 2^{n-2} + n \cdot 2^{n-1}$

$= \left(2^{n-1}\right)\left(\frac{n(n-1)}{2} + n\right)$

$= 2^{n-2} \cdot (n^2 + n)$

**Q.** **$1.4 \cdot {}^{35}c_2 - 3.7 \cdot {}^{35}c_3 + 5.10 \, {}^{35}c_4 \ldots$**

**Sol.** $= \sum_{r=2}^{35} (2r-3)(3r-2) \, {}^{35}C_r (-1)^r$

put r = 0

$\Rightarrow$ $(-3)(-2) \, {}^{35}C_0 = 6$

Now put r = 1

$\Rightarrow (-1)(-1)(1) \, {}^{35}C_1 = -1 \times 35 \times -1 = 35$

Add both these to above and subtract

$= \sum_{r=0}^{35} (6r^2 - 13r + 6) \, {}^{35}C_r (-1)^r - 41$

$= 6\sum_{r=0}^{35} r^2 \cdot {}^{35}C_r (-1)^r - 13\sum_{r=0}^{35} r \cdot {}^{35}C_r (-1)^r + 6\sum_{r=0}^{35} {}^{35}C_r (-1)^r - 41$

$= -41$

**Q.** $1 \cdot 3 \cdot {}^{20}C_1 + 3 \cdot 7 \cdot {}^{20}C_2 + 5 \cdot 11 \cdot {}^{20}C_3 + \ldots$

**Sol.** $T_r = (2r-1)(4r-1)\,{}^{20}C_r$

$$= \sum_{r=1}^{20} (2r-1)(4r-1)\,{}^{20}C_r$$

$$= \sum_{r=1}^{20} (8r^2 - 6r + 1)\,{}^{20}C_r$$

$$= 8\sum_{r=1}^{20} r^2 \cdot {}^{20}C_r - 6\sum_{r=1}^{20} r \cdot {}^{20}C_r + \sum_{r=1}^{20} {}^{20}C_r$$

$$= 8 \cdot 20 \cdot 21 \cdot 2^{18} - 6 \cdot 20 \cdot 2^{20-1} + 2^{20} - 1$$

**Q.** **Show that:** $S = C_0 + 3.C_1 + 5.C_2 + \ldots + (2n+1)C_n = (n+1).2^n$

**Sol.** $S = 1.\,C_0 + 3.C_1 + 5.C_2 + \ldots + (2n+1)\,C_n$

$+ S = (2n+1)\,C_n + (2n-1)\,C_{n-1} + \ldots + 1.C_0$

$2S = (2n+2)\,[C_0 + C_1 + \ldots + C_n]$

$$S = \frac{2(n+1)}{2} \times 2^n$$

$\therefore\ S = (n+1)2^n.$

**Q.** **Find** $S = \frac{1}{2}\,{}^{10}C_0 - {}^{10}C_1 + {}^{10}C_2 \cdot 2 - {}^{10}C_3 \cdot 2^2 + \ldots + {}^{10}C_{10}\,2^9$

**Sol.** $S = \frac{1}{2}[{}^{10}C_0 - {}^{10}C_1 2^1 + {}^{10}C_2 2^2 + \ldots + {}^{10}C_{10}\,2^{10}]$

$$S = \frac{1}{2}[1-2]^{10} = \frac{1}{2}$$

**Q.** **Show that** $C_0^2 + C_1^2 + C_2^2 + C_3^2 + \ldots + C_n^2 = {}^{2n}C_n$

**Sol.** **(M 1)**

By method of selection.

Consider n boys and n girls in a class. Total number of ways to select n students out of 2n is ${}^{2n}C_n$. Alternatively, the same selection can be done if out of n boys, zero boys and out of n girls, n girls are selected or out of n boys, 1 boy and out of n girls, (n–1) girls are selected and so on...

Hence $^{2n}C_n = {}^nC_0\,{}^nC_n + {}^nC_1\,{}^nC_{n-1} + {}^nC_2\,{}^nC_{n-2} + \ldots$

$= {}^nC_0\,{}^nC_0 + {}^nC_1\,{}^nC_1 + {}^nC_2\,{}^nC_2 + \ldots$

$\therefore \quad {}^{2n}C_n = C_0^2 + C_1^2 + C_2^2 + \ldots$

**(M 2)**

Multiplication of two series

$(1 + x)^n = C_0 + C_1x + C_2x^2 + \ldots$

Replace x by $\frac{1}{x}$

$$\left(1+\frac{1}{x}\right)^n = C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + \ldots$$

$$(1+x)^n\left(1+\frac{1}{x}\right)^n = \left(C_0 + C_1x + C_2x^2 + \ldots\right)\left(C_0 + \frac{C_1}{x} + \frac{C_2}{x_2} + \ldots\right)$$

$$\frac{(1+x)^{2n}}{x^n} = C_0\left\{C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + \ldots\right\} + C_1x\left\{C_0 + \frac{C_1}{x} + \frac{C_2}{x_2} + \ldots\right\} + C_2x^2\left\{C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + \ldots\right\} + \ldots$$

Taking coefficient of $x^0$ from both side

coefficient of $x^n$ in $(1 + x)^{2n} = C_0^2 + C_1^2 + C_2^2 + \ldots + C_n^2$

$$^{2n}C_n = C_0^2 + C_1^2 + C_2^2 + \ldots + C_n^2$$

**Q.** **Find S where** $S = 1.C_0^2 + 2.C_1^2 + 3.C_2^2 + \ldots + (n+1)C_n^2$

**Sol.**

$$S = 1.C_0^2 + 2.C_1^2 + 3.C_2^2 + \ldots + (n+1)C_n^2$$

$$+S = (n+1)C_n^2 + nC_{n-1}^2 + \ldots + 1.C_0^2$$

---

$$2S = (n+2)C_0^2 + (n+2)C_1^2 + \ldots + (n+2)C_n^2$$

$$S = \frac{(n+2)}{2} \times {}^{2n}C_n$$

**Q.** **Show that** $C_0 + 2C_1 + 3C_2 + 4C_3 + \ldots + (n+1)C_n = (n+2) \cdot 2^{n-1}$

**Sol.**

**(M 1)**

$S = C_0 + 2C_1 + 3C_2 + 4C_3 + \ldots + (n+1)C_n$

$+ S = (n+1)C_n + nC_{n-1} + \ldots + 2C_1 + C_0$

$2S = (n+2)[C_0 + C_1 + \ldots + C_n]$

$$S = \frac{n+2}{2}\,[2^n] = (n+2)\,2^{n-1}$$

**(M 2)**

**Consider**

$(1+x)^n = C_0 + C_1x + C_2x^2 + \ldots + C_nx^n$

$$\frac{d}{dx}\left[(1+x)^n x\right] = \frac{d}{dx}\,[C_0x + C_1x^2 + C_2x^3 + \ldots + C_nx^{n+1}]$$

$(1+x)^n + nx(1+x)^{n-1} = C_0 + 2C_1x + 3C_2x^2 + \ldots + (n+1)C_nx^n$

Put $x = 1$

$2^n + n2^{n-1} = C_0 + 2C_1 + 3C_2 + \ldots + (n+1)C_n$

$\therefore \quad C_0 + 2C_1 + 3C_2 + \ldots + (n+1)C_n = (n+2)\,2^{n-1}$

**Q.** **Show that $S = 1.C_1 + 2.C_2 + 3.C_3 + \ldots + n.C_n = n.2^{n-1}$**

**Sol.** **(M 1)**

**Consider**

$(1+x)^n = C_0 + C_1x + C_2x^2 + \ldots + C_nx^n$

Differentiate both sides with respect to x

$n(1+x)^{n-1} = C_1 + 2C_2x + \ldots + n.C_nx^{n-1}$

Put $x = 1$

$n(2)^{n-1} = C_1 + 2C_2 + 3C_3 + \ldots + n.C_n$

**(M 2)**

$T_r = r.{}^nC_r$

$$= n.{}^{n-1}C_{r-1}\left[{}^nC_r = \frac{n}{r}\left({}^{n-1}C_{r-1}\right)\right]$$

$$S = \sum_{r=1}^{n} n.\ {}^{n-1}C_{r-1} = n\sum_{r=1}^{n} {}^{n-1}C_{r-1}$$

$= n.(2^{n-1})$

**Q.** **For any positive integers m, n (with $n \geq m$) let $\binom{n}{m} = {}^nC_m$.**

**Prove that $\binom{n}{m} + \binom{n-1}{m} + \binom{n-2}{m} + \ldots + \binom{m}{m} = \binom{n+1}{m+1}$**

**Sol.** **(M 1)**

${}^mC_m + {}^{m+1}C_m + \ldots + {}^nC_m$

$= ({}^{m+1}C_{m+1} + {}^{m+1}C_m) + {}^{m+2}C_m + \ldots + {}^nC_m \quad [\because {}^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1}]$

$= ({}^{m+2}C_{m+1} + {}^{m+2}C_m) + {}^{m+3}C_m + \ldots + {}^{n}C_m$

$= ({}^{m+3}C_{m+1} + {}^{m+3}C_m) + \ldots + {}^{n}C_m$

$= {}^{n+1}C_{m+1}$

**(M 2)**

Coefficient of $x^m$ in $(1+x)^m + (1+x)^{m+1} + (1+x)^{m+2} + \ldots + (1+x)^n$

Coefficient of $x^m$ in $(1+x)^m \dfrac{[(1+x)^{n-m+1} - 1]}{(1+x)-1}$

Coefficient of $x^{m+1}$ in $(1+x)^{m+n-m+1} = {}^{n+1}C_{m+1}$

**Q.** **Show that $\sum_{k=0}^{n} {}^{n}C_k \sin kx \cos(n-k)x = 2^{n-1}\sin nx$**

**Sol.** Let $s = C_0 \sin 0 \cos nx + C_1 \sin x \cos(n-1)x + \ldots + {}^{n}C_n \sin(nx)\cos(ox)$

$s = C_n \sin(nx)\cos(ox) + C_{n-1}\sin(n-x)x \cos x + \ldots + C_0 \sin 0 \cos nx$

+

$2s = C_0(\sin nx) + C_1(\sin nx) + \ldots + C_n \sin(nx)$

$s = \dfrac{(C_0 + C_1 + \ldots + C_n)}{2}\sin nx = \dfrac{2^n}{2}.\sin nx$

## Use of complex number in the binomial theorem

In the binomial expression of $(\cos\theta + i\sin\theta)^n$ equate the real part to $\cos n\theta$ and the imaginary part to $\sin n\theta$, we get

$\cos n\theta = \cos^n\theta - {}^{n}C_2 \cos^{n-2}\theta \sin^2\theta + {}^{n}C_4 \cos^{n-4}\theta \sin^4\theta + \ldots$ ...(i)

$\sin n\theta = {}^{n}C_1 \cos^{n-1}\theta \sin\theta - {}^{n}C_3 \cos^{n-3}\theta \sin^3\theta + {}^{n}C_5 \cos^{n-5}\theta \sin^5\theta + \ldots$ ...(ii)

$\Rightarrow$ Dividing (i) and (ii)

$$\tan n\theta = \frac{{}^{n}C_1 \tan\theta - {}^{n}C_3 \tan^3\theta + {}^{n}C_5 \tan^5\theta - {}^{n}C_7 \tan^7\theta + \cdots}{1 - {}^{n}C_2 \tan^2\theta + {}^{n}C_4 \tan^4\theta - {}^{n}C_6 \tan^6\theta + \cdots}$$

**Q.** **Find the sum ${}^{n}C_0 - {}^{n}C_2 + {}^{n}C_4 - {}^{n}C_6 + \ldots {}^{n}C_r$.**

**Sol.** $(1+x)^n = {}^{n}C_0 + {}^{n}C_1 x + {}^{n}C_2 x + \ldots + {}^{n}C_n x^n$

Put $x = i$ Then,

$(1+i)^n = {}^{n}C_0 + {}^{n}C_1 i + {}^{n}C_2 i^2 + {}^{n}C_3 i^3 + {}^{n}C_4 i^4 + \ldots$

$= ({}^{n}C_0 - {}^{n}C_2 + {}^{n}C_4 - {}^{n}C_6 + \ldots) + i({}^{n}C_1 - {}^{n}C_3 + {}^{n}C_5 + \ldots)$

$\Rightarrow$ ${}^nC_0 - {}^nC_2 + {}^nC_4 - {}^nC_6 + \ldots$ = Real part of $(1 + i)^n$

$$= \text{Re}\left[(\sqrt{2})^n \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right)^n\right]$$

$$= \text{Re}\left[2^{n/2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)^n\right]$$

$$= \text{Re}\left[2^{n/2}\left(\cos\frac{n\pi}{4} + i\sin\frac{n\pi}{4}\right)\right] \text{ [De' moivre's theorem]}$$

$$= 2^{n/2}\cos\frac{n\pi}{4}$$

Also,

$${}^nC_1 - {}^nC_3 + {}^nC_5 - \cdots = \text{Im}\left[(1+i)^n\right] = 2^{n/2}\sin\left(\frac{n\pi}{4}\right)$$

**Q.** **Find the value of ${}^{4n}C_0 + {}^{4n}C_4 + {}^{4n}C_8 + \ldots + {}^{4n}C_{4n}$.**

**Sol.** ${}^{4n}C_0 + {}^{4n}C_2x^2 + {}^{4n}C_4x^4 + \ldots + {}^{4n}C_{4n}x^{4n}$

$$= \frac{1}{2}\left[(1+x)^{4n} + (1-x)^{4n}\right]$$

Putting x = 1 and x = i, we get

$${}^{4n}C_0 + {}^{4n}C_2 + {}^{4n}C_4 + \ldots + {}^{4n}C_{4n} = \frac{1}{2}\left[2^{4n}\right] \quad \ldots(i)$$

and ${}^{4n}C_0 - {}^{4n}C_2 + {}^{4n}C_4 - \ldots + {}^{4n}C_{4n}$ $\quad \ldots(ii)$

$$= \frac{1}{2}\left[(1+i)^{4n} + (1-i)^{4n}\right]$$

Add (i) and (ii)

$$2\left[{}^{4n}C_0 + {}^{4n}C_4 + \ldots + {}^{4n}C_{4n}\right] = 2^{4n-1} + \frac{1}{2}\left[(1+i)^{4n} + (1-i)^{4n}\right]$$

Now,

$(1 + i)^{4n} + (1 - i)^{4n}$

$$= \left[\sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)\right]^{4n} + \left[\sqrt{2}\left(\cos\frac{\pi}{4} - i\sin\frac{\pi}{4}\right)\right]^{4n}$$

$= 2^{2n}(\cos n\pi + i\sin n\pi) + 2^{2n}(\cos n\pi - i\sin n\pi)$

$= 2^{2n+1}\cos n\pi = 2^{2n+1}(-1)^n$

$\therefore 2[{}^{4n}C_0 + {}^{4n}C_4 + \ldots + {}^{4n}C_{4n}] = 2^{4n-1} + \frac{1}{2}\,2^{2n+1}(-1)^n$

$\Rightarrow {}^{4n}C_0 + {}^{4n}C_4 + \ldots + {}^{4n}C_{4n} = 2^{4n-2} + (-1)^n\,2^{2n-1}$

## Product Of Two Binomial

**Q.** **Show that $C_0C_1 + C_1C_2 + C_2C_3 + \ldots + C_{n-1}C_n = {}^{2n}C_{n+1}$**

**Sol.** **(M-1)**

$(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \ldots$

$$\left(1+\frac{1}{x}\right)^n = C_0 + C_1\left(\frac{1}{x}\right) + C_2\left(\frac{1}{x^2}\right) + C_3\left(\frac{1}{x^3}\right) + \ldots$$

$$(1+x)^n\left(1+\frac{1}{x}\right)^n = \left(C_0 + C_1x + C_2x^2 + \ldots.\right)\left(C_0 + \frac{C_1}{x} + \frac{C_2}{x^2} + \ldots\right)$$

Equating coefficients of $\frac{1}{x}$ both sides

Coefficient of $\frac{1}{x}$ in $(1+x)^n \dfrac{(1+x)^n}{x^n} = C_0C_1 + C_1C_2 + C_2C_3 + \ldots + C_{n-1}C_n$

Coefficient of $x^{n-1}$ in $(1+x)^{2n} = C_0C_1 + C_1C_2 + C_2C_3 + \ldots {}^{2n}C_{n-1} = {}^{2n}C_{n+1}$

So, $C_0C_1 + C_1C_2 + C_2C_3 + \ldots = {}^{2n}C_{n+1}$

**(M-2)**

LHS $= C_0C_{n-1} + C_1C_{n-2} + C_2C_{n-3} + \ldots + C_{n-1}C_0$ consider 'n' boys and 'n' girls out of which 'n−1' students are to be selected. First way is to select zero boys out of n and (n − 1) girls out of n. For this number of ways are ${}^nC_0 \cdot {}^nC_{n-1}$ second ways is to select 1 boy out of n and (n − 2) girls out of n. This can be done in ${}^nC_1\,{}^nC_{n-2}$ ways and so on...

Hence total number of ways equals L.H.S. Now, another ways is to directly select (n − 1) students out of '2n' in ${}^{2n}C_{n-1}$ ways which is equal to ${}^{2n}C_{n+1}$ (R.H.S)

Hence, $C_0C_1 + C_1C_2 + \ldots = {}^{2n}C_{n+1}$

**Q.** For r = 0, 1, ...,10 Let $A_r$, $B_r$ and $C_r$ denote respectively, the coefficient of $x^r$ in the expansion of $(1+x)^{10}$, $(1+x)^{20}$ and $(1+x)^{30}$. Then $\sum_{r=1}^{10} A_r\left(B_{10}B_r - C_{10}.Ar\right)$ is equal to

(A) $B_{10} - C_{10}$ (B) $A_{10}(B_{10}^2 - C_{10}A_{10})$ (C) 0 (D) $C_{10} - B_{10}$

**Sol.** (D)

$A_r = {}^{10}C_r$, $B_r = {}^{20}C_r$, $C_r = {}^{30}C_r$

$$B_{10}\sum_{r=1}^{10}({}^{10}C_r)({}^{20}C_r) - C_{10}\sum_{r=1}^{10}({}^{10}C_r)^2$$

$${}^{20}C_{10}\left(\sum_{r=0}^{10}{}^{10}C_{10-r}\,{}^{20}C_r\right) - {}^{30}C_{10}\left(\sum_{r=0}^{10}({}^{10}C_r)^2\right) - {}^{20}C_{10} + {}^{30}C_{10}$$

$${}^{20}C_{10}\,{}^{30}C_{10} - {}^{30}C_{10}\,{}^{20}C_{10} + {}^{30}C_{10} - {}^{20}C_{10}$$

$${}^{30}C_{10} - {}^{20}C_{10} = C_{10} - B_{10}$$

**Q.** The value of r for which ${}^{20}C_r\,{}^{20}C_0 + {}^{20}C_{r-1}\,{}^{20}C_1 + {}^{20}C_{r-2}\,{}^{20}C_2 + ... + {}^{20}C_0\,{}^{20}C_r$ is maximum is

(A) 15 (B) 20 (C) 11 (D) 10

**Sol.** (B)

Out of 20 boys and 20 girls, total 'r' students are selected. This is stated by the given series.

Hence it equals ${}^{(20+20)}C_r = {}^{40}C_r$

${}^{40}C_r$ is maximum at $r = \frac{40}{2} = 20$

**Q.** Show that : ${}^{2n-2}C_{n-2} + 2.{}^{2n-2}C_{n-1} + {}^{2n-2}C_n > \frac{4n}{n+1}$, $n \in N$ $n > 2$

**Sol.** ${}^{2n-2}C_{n-2} \geq 1$, $2.{}^{2n-2}C_{n-1} \geq 2$, ${}^{2n-2}C_n \geq 1$

Adding, LHS $\geq 4$

Now, RHS $= \frac{4n+4-4}{n+1} = 4 - \frac{4}{n+1} < 4$

$\Rightarrow$ RHS < LHS.

**Q.** **Prove that:** $2.C_0 + \frac{2^2}{2}.C_1 + \frac{2^3}{3}C_2 + \frac{2^4}{4}C_3 + ... + \frac{2^{n+1}}{n+1}C_n = \frac{3^{n+1}-1}{n+1}$

**Sol.** **(M 1)**

$$\int_0^2 (1+x)^n \, dx = \int_0^2 \left(C_0 + C_1x + C_2x^2 + ... + C_nx^n\right) dx$$

$$\left.\frac{(1+x)^{n+1}}{n+1}\right|_0^2 = \left[C_0x + \frac{C_1x^2}{2} + C_2\frac{x^3}{3} + ... + C_n\frac{x^{n+1}}{n+1}\right]$$

$$\frac{3^{n+1}-1}{n+1} = 2.C_0 + \frac{C_1.2^2}{2} + C_2.\frac{2^3}{3} + ....$$

**(M 2)**

$$T_r = \frac{2^{r+1}\,{}^nC_r}{r+1}$$

$$S = \sum_{r=0}^{n} \frac{2^{r+1}\,{}^nC_r}{r+1} = \sum_{r=0}^{n} \frac{2^{r+1}\,{}^{n+1}C_{r+1}}{(n+1)}$$

$$= \frac{1}{n+1}\sum_{r=0}^{n} {}^{n+1}C_{r+1}2^{r+1}$$

$$= \frac{1}{n+1}\sum_{r=0}^{n} {}^{n+1}C_{r+1}2^{r+1}$$

$$= \frac{1}{n+1}\left[1 + {}^{n+1}C_1 2^1 + {}^{n+1}C_2 2^2 + {}^{n+1}C_3 2^3 + ... + {}^{n+1}C_{n+1}2^{n+1} - 1\right]$$

$$= \frac{1}{n+1}\left[(1+2)^{n+1} - 1\right]$$

$$= \frac{3^{n+1}-1}{n+1}$$

**Q.** **Prove :** $\frac{C_0}{2} + \frac{C_1}{3} + \frac{C_2}{4} + \frac{C_3}{5} + ... + \frac{C_n}{n+2} = \frac{1+n.2^{n+1}}{(n+1)(n+2)}$

**Sol.** $(1+x)^n = C_0 + C_1x + C_2x^2 + ...$

$$\int_0^1 x(1+x)^n \, dx = \int_0^1 \left(C_0x + C_1x^2 + C_2x^3 + ...\right) dx$$

Put $(1 + x) = t$

$dx = dt$

$$\int_1^2 (t-1)t^n dt = \left( C_0 \frac{x^2}{2} + C_1 \frac{x^3}{3} + C_2 \frac{x^4}{4} + \ldots \right)_0^1$$

$$\int_1^2 t^{n+1} dt - \int_1^2 t^n dt = \left( \frac{C_0}{2} + \frac{C_1}{3} + \frac{C_2}{4} + \ldots \frac{C_n}{n+2} \right)$$

$$\left[ \frac{t^{n+2}}{n+2} \right]_1^2 - \left[ \frac{t^{n+1}}{n+1} \right]_1^2 = \frac{C_0}{2} + \frac{C_1}{3} + \frac{C_2}{4} + \ldots + \frac{C_n}{n+2}$$

$$\frac{1}{n+2}\left[2^{n+2} - 1\right] - \frac{1}{n+1}\left(2^{n+1} - 1\right) = \frac{C_0}{2} + \frac{C_1}{3} + \frac{C_2}{4} + \ldots$$

$$\therefore \frac{C_0}{2} + \frac{C_1}{3} + \frac{C_2}{4} + \ldots = 2^{n+1}\left[\frac{2}{n+2} - \frac{1}{n+1}\right] - \frac{1}{n+2} + \frac{1}{n+1}$$

$$= 2^{n+1} \frac{(2n+2-n-2)}{(n+2)(n+1)} + \frac{(n+2-n-1)}{(n+1)(n+2)}$$

$$= \frac{2^{n+1}.n+1}{(n+1)(n+2)}$$

**Q.** **Let $S_n = 1 + q + q^2 + \ldots + q^n$ and $T_n = 1 + \left(\frac{q+1}{2}\right) + \left(\frac{q+1}{2}\right)^2 + \ldots + \left(\frac{q+1}{2}\right)^n$ where q is a real number and $q \neq 1$.**
**If $^{101}C_1 + {}^{101}C_2 . S_1 + \ldots + {}^{101}C_{101} . S_{100} = \alpha . T_{100}$ then $\alpha$ is**
**(A) $2^{99}$ (B) 202 (C) 200 (D) $2^{100}$**

**Sol.** **(D)**

$$S_n = \frac{q^{n+1} - 1}{q - 1}; \; T_n = \frac{\left(\frac{q+1}{2}\right)^{n+1} - 1}{\frac{q+1}{2} - 1} = \frac{(q+1)^{n+1} - 2^{n+1}}{(q-1)2^n}$$

Now $$^{101}C_1 + {}^{101}C_2 \frac{q^2 - 1}{q - 1} + {}^{101}C_3 \frac{q^3 - 1}{q - 1} + \ldots + {}^{101}C_{101}\left(\frac{q^{101} - 1}{q - 1}\right)$$

$$= \frac{1}{q-1}\left[\left({}^{101}C_1 q^1 + {}^{101}C_2 q^2 + \ldots + {}^{101}C_{101} q^{101}\right) - \left({}^{101}C_1 + {}^{101}C_2 + \ldots + {}^{101}C_{101}\right)\right]$$

$$= \frac{1}{(q-1)}\left\{\left[(1+q)^{101} - 1\right] - \left[2^{101} - 1\right]\right\} = \alpha . T_{100}$$

$$\Rightarrow \frac{(1+q)^{101}-2^{101}}{(q-1)} = \alpha \frac{(q+1)^{101}-2^{101}}{(q-1)2^{100}}$$

$\alpha = 2^{100}$

**Q.** $\begin{vmatrix} \sum_{k=0}^{n} k & \sum_{k=0}^{n} {}^nC_k \cdot k^2 \\ \sum_{k=0}^{n} k.{}^nC_k & \sum_{k=0}^{n} {}^nC_k.3^k \end{vmatrix} = 0$ **holds for some positive integer n then find** $\sum_{k=0}^{n} \frac{{}^nC_k}{k+1}$.

**Sol.** $\begin{vmatrix} \frac{n(n+1)}{2} & n(n+1)2^{n-2} \\ n.2^{n-1} & (1+3)^n \end{vmatrix} = 0$

$$n(n+1)\begin{vmatrix} \frac{1}{2} & 2^{n-2} \\ n.2^{n-1} & 2^n.2^n \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & 2^{n-1} \\ n & 2^{n+1} \end{vmatrix} = 0$$

$2^{n+1} - n.2^{n-1} = 0 \Rightarrow (4-n)\,2^{n-1} = 0$

$n = 4$

$$\therefore \sum_{k=0}^{4} \frac{{}^4C_k}{k+1} = {}^4C_0 + \frac{{}^4C_1}{2} + \frac{{}^4C_2}{3} + \frac{{}^4C_3}{4} + \frac{{}^4C_4}{5}$$

$$= 1 + 2 + 2 + 1 + \frac{1}{5} = 6.2$$

**Q.** **Prove :** $\left(\frac{C_0 + C_1}{C_0}\right)\left(\frac{C_1 + C_2}{C_1}\right)\left(\frac{C_2 + C_3}{C_2}\right)\cdots\left(\frac{C_{n-1} + C_n}{C_{n-1}}\right) = \frac{(n+1)^n}{n!}$

**Sol.** LHS : $\prod_{r=0}^{n-1}\left(\frac{C_r + C_{r+1}}{C_r}\right) = \prod_{r=0}^{n-1} \frac{{}^{n+1}C_{r+1}}{{}^nC_r}$

$$= \prod_{r=0}^{n-1} \frac{n+1}{r+1} = (n+1)^n \times \frac{1}{1.2.3.....n} = \frac{(n+1)^n}{n!}$$

**Q.** $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \ldots + a_{2n}x^{2n}$

**(i) Show that $a_r = a_{2n-r}$ $\forall$ r $\in$ W**

**Sol.** Replace x by $\frac{1}{x}$

$$\left(1+\frac{1}{x}+\frac{1}{x^2}\right)^n = a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \ldots + \frac{a_{2n}}{x^{2n}}$$

Multiply both sides by $x^{2n}$

$(1 + x + x^2)^n = a_0x^{2n} + a_1x^{2n-1} + a_2x^{2n-2} + \ldots + a_{2n}$

Also, $(1 + x + x^2)^n = a_0x^0 + a_1x^1 + a_2x^2 + \ldots + a_{2n}x^{2n}$ ...(A)

comparing coefficient of $x^r$ both sides $a_r = a_{2n-r}$

**(ii) $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \ldots + a_{2n}^2 = a_n$**

**Sol.** Replaced x by $-\frac{1}{x}$

$$\left(1-\frac{1}{x}+\frac{1}{x^2}\right)^n = a_0 - \frac{a_1}{x} + \frac{a_2}{x^2} \ldots \quad \ldots (A)$$

$(x^2 - x + 1)^n = a_0x^{2n} - a_1x^{2n-1} + a_2x^{2n-2} \ldots$ ...(B)

Multiply (A) and (B)

$(1 + x + x^2)^n (x^2 - x + 1)^n = (a_0 + a_1x + a_2x^2 + \ldots)(a_0x^{2n} - a_1x^{2n-1} + a_2x^{2n-2} - \ldots)$

$(x^4 + x^2 + 1)^n = a_0\{a_0x^{2n} - a_1x^{2n-1} + \ldots\} + a_1x\{a_0x^{2n} - a_1x^{2n-1} + a_2x^{2n-2} + \ldots\} +$

$a_2x^2\{a_0x^{2n} - a_1x^{2n-1} + a_2x^{2n-2} + \ldots\} + \ldots$ ...(C)

Taking coefficient of $x^{2n}$ from RHS-

$a_0^2 - a_1^2 + a_2^2 - a_3^2 + \ldots + a_{2n}^2$ ...(D)

Replace x by $x^2$ in (A)

$(1 + x^2 + x^4)^n = a_0 + a_1x^2 + a_2x^4 + \ldots + a_nx^{2n} + \ldots + a_{2n}x^{4n}$ ...(E)

coefficient of $x^{2n}$ in RHS is $a_n$ ...(F)

from (F) and (D), $a_0^2 - a_1^2 + a_2^2 - a_3^2 + \ldots + a_{2n}^2 = a_n$.

**(iii) $a_0a_1 - a_1a_2 + a_2a_3 \ldots = 0$**

**Sol.** Taking coefficient of $x^{2n-1}$ from RHS in (C) & from RHS in (E)

$-a_0a_1 + a_1a_2 - a_2a_3 + \ldots = 0$

$a_0a_1 - a_1a_2 + a_2a_3 - \ldots = 0$

**(iv) $a_0a_2 - a_1a_3 + a_2a_4 - \ldots =$**

**Sol.** Taking coefficient of $x^{2n-2}$ from RHS in (C) and (E)

$a_0a_2 - a_1a_3 + a_2a_4 - \ldots = a_{n-1}$

**Note :** $a_{n-1} = a_{n+1}$ $(\because a_r = a_{2n-r})$

**Q.** **If $(1 + x)^n = C_0 + C_1x + C_2x^2 + \ldots + C_nx^n$, then show that the sum of the products of the $C_i$'s taken two at a time, represented by $\sum_{0\le} \sum_{i<j\le n} C_iC_j$ is equal to $2^{2n-1} - \frac{2n!}{2(n!)^2}$**

**Sol.** To find : $C_0(C_1 + C_2 + \ldots + C_n) +$ $\quad i = 0$

$C_1(C_2 + C_3 + \ldots + C_n) +$ $\quad i = 1$

$C_2(C_3 + \ldots + C_n) +$ $\quad i = 2$

$\vdots$ $\quad \vdots$

$\vdots$ $\quad \vdots$

$\vdots$ $\quad \vdots$

$\vdots$ $\quad \vdots$

$C_{n-1}C_n$ $\quad i = n - 1$

$$(C_0 + C_1 + C_2 + \ldots + C_n)^2 = \sum_{r=0}^{n} C_r^2 + 2\sum_{0\le i<j\le n} C_iC_j$$

$$\frac{(2^n)^2 - {}^{2n}C_n}{2} = \sum_{0\le i<j\le n}\sum C_iC_j$$

$$\therefore \sum_{0\le i<j\le n}\sum C_iC_j = 2^{2n-1} - \frac{(2n)!}{2(n!)^2}$$

**Q.** **Prove that** $2^k\binom{n}{0}\binom{n}{k} - 2^{k-1}\binom{n}{1}\binom{n-1}{k-1} + 2^{k-2}\binom{n}{2}\binom{n-2}{k-2} \ldots\ldots (-1)^k\binom{n}{k}\binom{n-k}{0} = \binom{n}{k}$

**Sol.** $$\text{LHS} = \sum_{r=0}^{k}(-1)^r 2^{k-r}\ \underbrace{{}^nC_r\ \ {}^{n-r}C_{k-r}}$$

$$\downarrow$$

$$\frac{n!}{r!(n-r)!}\frac{(n-r)!}{(k-r)!(n-k)!} \times \frac{k!}{k!}$$

$$\frac{n!}{(n-k)!\,k!} \times \frac{k!}{r!(k-r)!}$$

$$= {}^nC_k\ \ {}^kC_r$$

$$\text{LHS} = {}^{n}C_k \sum_{r=0}^{k} (-1)^r \, {}^{k}C_r \, 2^{k-r} = {}^{n}C_k (2-1)^k = {}^{n}C_k$$

$$= 2^{n-1} (\sin nx)$$

**Binomial theorem for any index :**

When n is a negative integer or a fraction then the expansion of a binomial is possible only when

**(i)** Its first term is 1, and

**(ii)** Its second term is numerically less than 1. Thus when $n \notin N$ and $|x| < 1$, then it states

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots + \frac{n(n-1)(n-r+1)}{r!}x^r + \ldots \infty \infty$$

**General Terms :**

$$T_{r+1} = \frac{n(n-1)(n-2)\ldots(n-r+1)}{r!} \cdot x^r$$

**Note :**

(i) In this expansion the coefficient of different terms can not be expressed as ${}^{n}C_0$, ${}^{n}C_1$, ${}^{n}C_2$, ... because n is not a positive integer.

(ii) In this case there are infinite terms in the expansion.

**Some Important Expansions :**

If $|x| < 1$ and $n \in Q$ but $n \notin N$, then

**(a)** $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots + \frac{n(n-1)\ldots(n-r+1)}{r!}x^r + \ldots$

**(b)** $(1-x)^n = 1 - nx + \frac{n(n-1)}{2!}x^2 - \frac{n(n-1)(n-2)}{3!}x^3 + \ldots + \frac{n(n-1)\ldots(n-r+1)}{r!}(-x)^r + \ldots$

**(c)** $(1-x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + \ldots + \frac{n(n+1)\ldots(n+r-1)}{r!}x^r + \ldots$

**(d)** $(1+x)^{-n} = 1 - nx + \frac{n(n+1)}{2!}x^2 - \frac{n(n+1)(n+2)}{3!}x^3 + \ldots + \frac{n(n+1)\ldots(n+r-1)}{r!}(-x)^r + \ldots$

By putting n = 1, 2, 3 in the above results (c) and (d),

we get the following results

**(e)** $(1 - x)^{-1} = 1 + x + x^2 + x^3 + \ldots + x^r + \ldots$

**General term :** $T_{r+1} = x^r$

**(f)** $(1 + x)^{-1} = 1 - x + x^2 - x^3 + \ldots + (-x)^r + \ldots$

**General term :** $T_{r+1} = (-x)^r$

**(g)** $(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \ldots + (r + 1)x^r + \ldots$

**General term :** $T_{r+1} = (r + 1)x^r$

**(h)** $(1 + x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \ldots + (r + 1)(-x)^r + \ldots$

**General term :** $T_{r+1} = (r + 1)(-x)^r$

**(i)** $(1 - x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \ldots + \dfrac{(r + 1)(r + 2)}{2!}x^r + \ldots$

**General term** $= \dfrac{(r + 1)(r + 2)}{2!}x^r$

**(j)** $(1 + x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \ldots + \dfrac{(r + 1)(r + 2)}{2!}(-x)^r + \ldots$

**General term** $= \dfrac{(r + 1)(r + 2)}{2!}(-x)^r$

**Q.** **If $|x| < 2 / 3$ then the fourth term in the expansion of $\left(1 + \frac{3}{2}x\right)^{1/2}$ is**

**(A)** $\dfrac{27}{128}x^3$ **(B)** $-\dfrac{27}{128}x^3$ **(C)** $\dfrac{81}{256}x^3$ **(D)** $-\dfrac{81}{256}x^3$

**Sol.** **(A)**

$$T_4 = \frac{1/2\left(\frac{1}{2} - 1\right)\left(\frac{1}{2} - 2\right)}{3!} \cdot \left(\frac{3x}{2}\right)^3 = \frac{27}{128}x^3$$

**Q.** **The term independent of x in the expansion of $\left(\dfrac{1 - x}{1 + x}\right)^2$ is**

**(A) 4** **(B) 3** **(C) 2** **(D) 1**

**Sol.** **(D)**

$(1 - x)^2 (1 + x)^{-2}$

$\Rightarrow$ $(1 - 2x + x^2)(1 - 2x + x^2 + \ldots)$

$\Rightarrow$ so term independent of x = 1.

**Q.** **The coefficient of $x^5$ in the expansion of $(1 - x)^{-6}$ is**
**(A) 1260** **(B) – 1260** **(C) – 252** **(D) 252**

**Sol.** **(D)**

$x^5$ occurs in $T_6$ of the expansion, so

$$T_6 = T_{5+1} = \frac{6.7.8.9.10}{5!}x^5 = 252x^5$$

$\therefore$ Coefficient of $x^5 = 252$

**Application of binomial theorem :**

**(i)** With the help of binomial theorem, we can find out the value of sq. root, cube root and $4^{th}$ root etc. of the given number upto any decimal places.

**Q.** **The value of cube root of 1001 upto five decimal places is**
**(A) 10.03333** **(B) 10.00333** **(C) 10.00033** **(D) None of these**

**Sol.** **(B)**

$$(1001)^{1/3} = (1000 + 1)^{1/3} = 10\left(1 + \frac{1}{1000}\right)^{1/3} = 10\left\{1 + \frac{1}{3}\cdot\frac{1}{1000} + \frac{\frac{1}{3}\left(\frac{1}{3} - 1\right)}{2!}\frac{1}{1000^2} + \ldots\right\}$$

$= 10\ \{1 + 0.0003333 - 0.00000011 + \ldots\}$
$= 10.0033$

**(ii) To find the sum of Infinite series :**

We can compare the givein infinite series with the expansion of $(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$ and by finding the value of x and n and putting in $(1 + x)^n$ the sum of series is determined.

**Q.** **The sum of $1 + \frac{1}{4} + \frac{1.3}{4.8} + \frac{1.3.5}{4.8.12} + \ldots\infty$ is**

**(A) $\sqrt{2}$** **(B) $\frac{1}{\sqrt{2}}$** **(C) $\sqrt{3}$** **(D) $2^{3/2}$**

**Sol.** **(A)**

Comparing with $1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$

$nx = 1/4$ ...(i)

and $\frac{n(n-1)x^2}{2!} = 1.3 / 4.8$

or $\frac{nx(nx-x)}{2!} = \frac{3}{32} \Rightarrow \frac{1}{4}\left(\frac{1}{4} - x\right) = \frac{3}{16}$ (by (i))

$\Rightarrow \left(\frac{1}{4} - x\right) = \frac{3}{4} \Rightarrow x = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2}$ ...(ii)

Putting the value of x in (i)

$n(-1/2) = 1 / 4 \quad \Rightarrow n = -1/2$

$\therefore$ sum of series $= (1 + x)^n$

$$= \left(1 - \frac{1}{2}\right)^{-1/2} = (1/2)^{-1/2} = \sqrt{2}$$

**(iii) Approximation :**

**Q.** **If x is so small so that its square and higher power can be neglected. Find the value of** $\frac{\left(1 + \frac{2x}{3}\right)^{-5} + (4 + 2x)^{1/2}}{(4 + x)^{3/2}}$.

**Sol.** **(D)**

$$\frac{\left(1 + \frac{2x}{3}\right)^{-5} + (4 + 2x)^{1/2}}{(4 + x)^{3/2}} = \frac{\left(1 - \frac{10x}{3}\right) + 2\left(1 + \frac{x}{4}\right)}{8\left(1 + \frac{3x}{8}\right)} = \frac{1}{8}\left(3 - \frac{10x}{3} + \frac{x}{2}\right)\left(1 + \frac{3x}{8}\right)^{-1}$$

$$= \frac{3}{8}\left(1 - \frac{17x}{18}\right)\left(1 - \frac{3x}{8}\right) = \frac{3}{8}\left(1 - \frac{17x}{18} - \frac{3x}{8}\right) = \frac{72 - 95x}{24 \times 8}$$